

Osprey 3.5 V6 high-pressure fuel pump wear causing low rail pressure under

Document: tsb-26-103

Area: bulletins

1 Condition

Affected Osprey 3.5 V6 vehicles may set DTC P0087 (Fuel Rail/System Pressure Too Low) during hard acceleration or sustained high-load operation. Drivers report hesitation, reduced power, and an illuminated MIL when the rail cannot maintain commanded pressure. Freeze-frame data typically shows a large negative deviation between commanded and actual rail pressure at high engine load. See Osprey Fuel Service Manual :: Fuel Pressure and Sensor Faults for the full description of this code.

2 Cause

The condition is caused by accelerated wear of the high-pressure fuel pump cam follower and plunger in part number HPFP-4120, which reduces the pump's ability to sustain rail pressure under load. As the HPFP-4120 internal components wear, peak delivery falls off while idle and light-load pressure remain normal, masking the fault until high demand. Pump construction and operating theory are described in Osprey Fuel Service Manual :: High-Pressure Fuel Pump (GDI) , within the broader Osprey Fuel Service Manual :: Fuel Delivery section.

3 Diagnostic Verification

Before replacing any component, verify the fault following PROC-FUEL-DIAG by commanding a pressure sweep and comparing actual rail pressure against the fuel rail pressure sensor, part number FRS-4180. If the FRS-4180 reads accurately but actual pressure falls short under load, the pump is confirmed as the failed component rather than the sensor. The full verification routine is documented in Osprey Fuel Service Manual :: Fuel Trim and Pressure Diagnosis Procedure , and sensor replacement, if the FRS-4180 is also out of range, is covered in Osprey Fuel Service Manual :: Fuel Rail Pressure Sensor Replacement .

4 Repair Procedure

Replace the high-pressure fuel pump HPFP-4120 following PROC-FUEL-PUMP, observing all fuel-system depressurization and cleanliness requirements. Install a new cam follower with the pump and torque the mounting fasteners in sequence to specification. After installation, prime the system, clear DTC P0087, and confirm rail pressure holds under a high-load verification drive. The complete procedure is in Osprey Fuel Service Manual :: Fuel Pump Replacement Procedure .